ECE 590
Neural-Network-Based Large Language Models

Tuesdays and Thursdays 1:25-2:40
Fitzpatrick Schiciano A 1464

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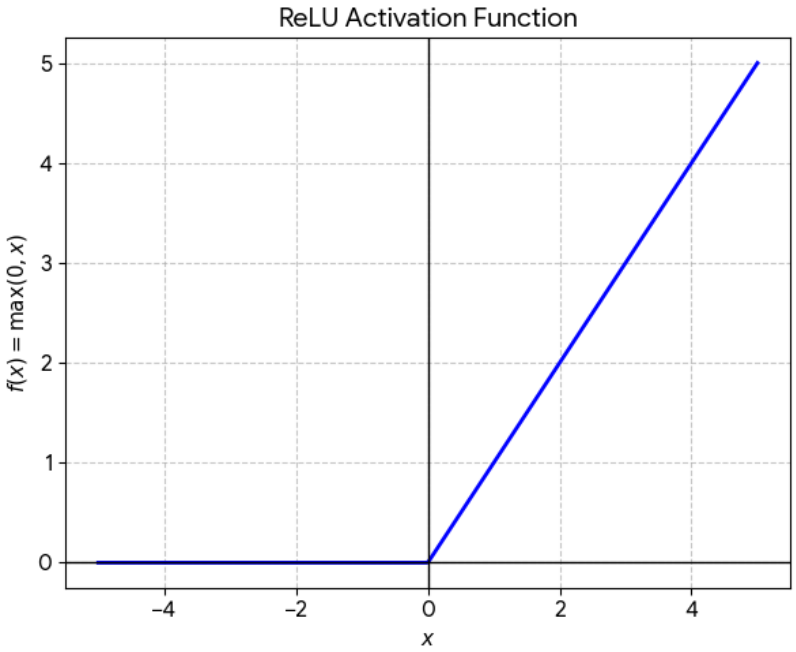
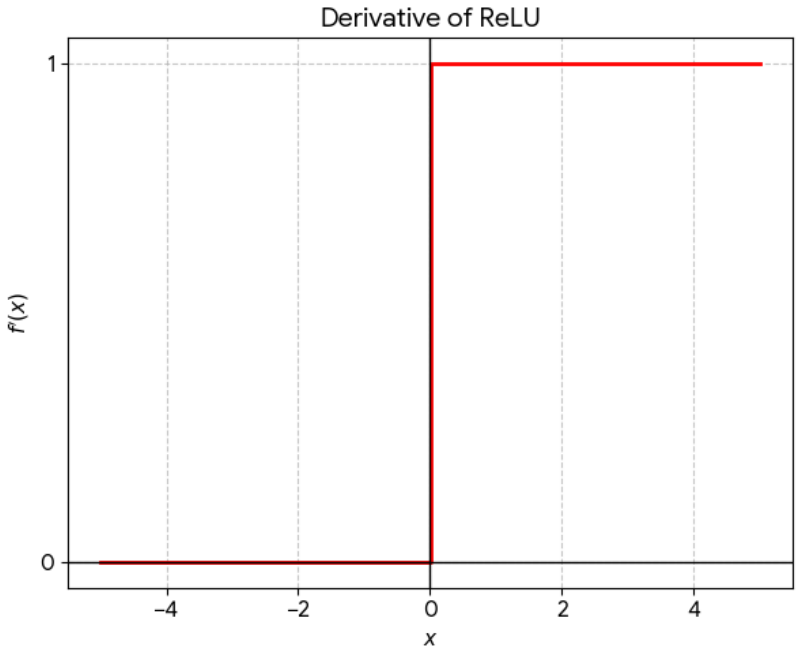
Neural-Network-Based Large Language Models

Lecture 5
01/29/2026

Skip Connections

- ReLu nonlinear units help address challenges of vanishing gradients, allowing deep NNs



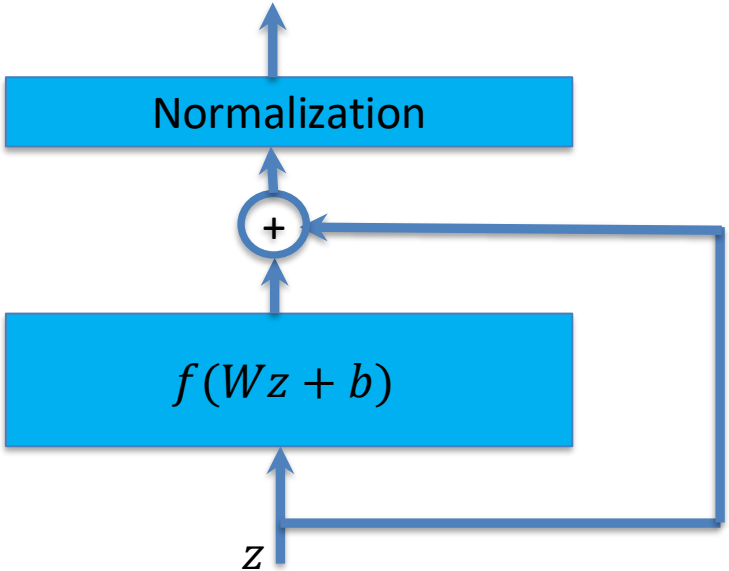


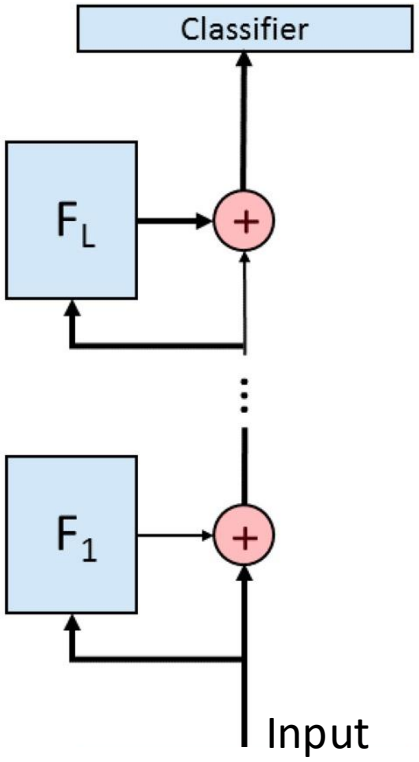
Skip Connections

- ReLu nonlinear units help address challenges of vanishing gradients, allowing deep NNs

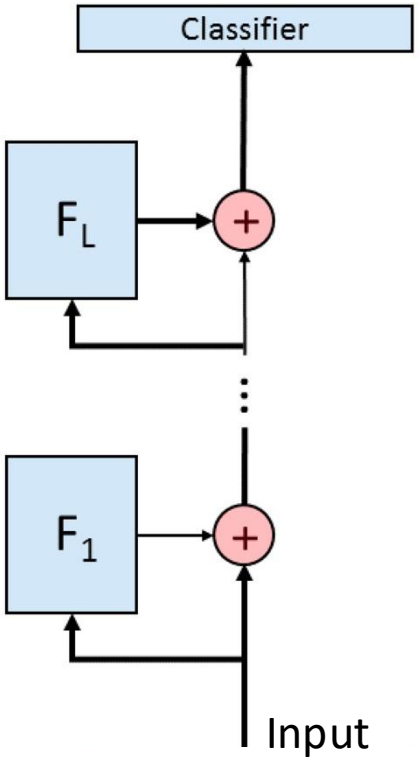
- Skip connections are another popular method used in deep networks

$f(\cdot)$
a pointwise
nonlinearity





Consider a Sequence of "Residual" Layers



- Consider the output of layer $i+1$ where $F(h_i)$ represents the role of the feedforward network with h_i output from layer below:

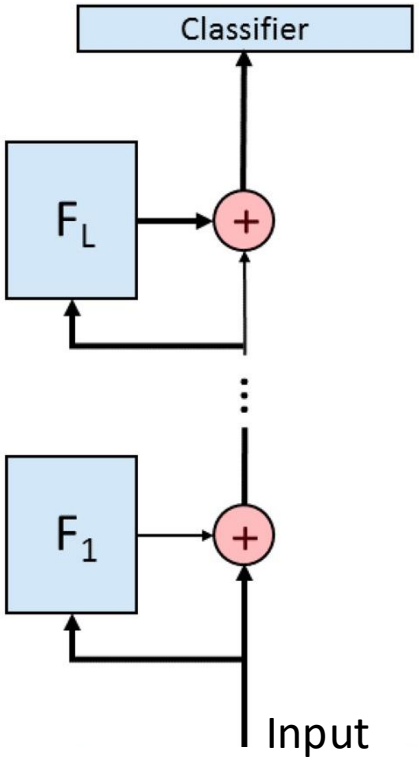
$$h_{i+1} = h_i + F(h_i)$$

- The loss $\ell(\cdot)$ is a function of output at layer L:

$$\ell(h_L) = \ell\left(h_{L-1} + F_{L-1}(h_{L-1})\right)$$

- Assume contribution of $F_{L-1}(h_{L-1})$ is small

- Perform Taylor series expansion about h_{L-1}



- Taylor expansion:

$$\ell(h_L) = \ell(h_{L-1}) + F_{L-1}(h_{L-1}) \cdot \nabla_{h_{L-1}} \ell(h_{L-1})$$
$$+O(F_{L-1}^2(h_{L-1}))$$

- We can similarly apply the same idea to $\ell(h_{L-1})$ and backwards to layer i

$$\ell(h_L) = \ell(h_i) + \sum_{j=i}^{L-1} F_j(h_j) \cdot \nabla_{h_j} \ell(h_j)$$
$$+O(F_j^2(h_j))$$

Implications

$$\ell(h_L) \approx \ell(h_i) + \sum_{j=i}^{L-1} F_j(h_j) \cdot \nabla_{h_j} \ell(h_j)$$

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$$\ell(h_L) \approx \ell(h_i) + \sum_{j=i}^{L-1} \underbrace{F_j(h_j) \cdot \nabla_{h_j} \ell(h_j)}$$

- Insight: To reduce loss, require $F_j(h_j) \propto -\nabla_{h_j} \ell(h_j)$

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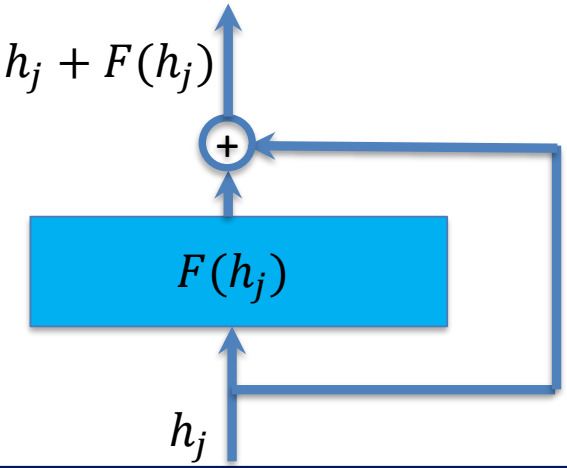
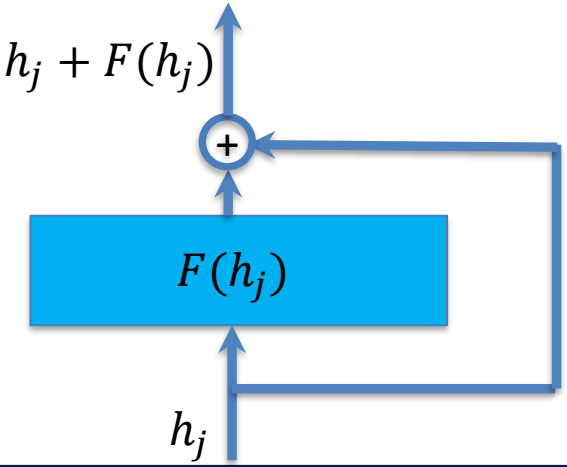


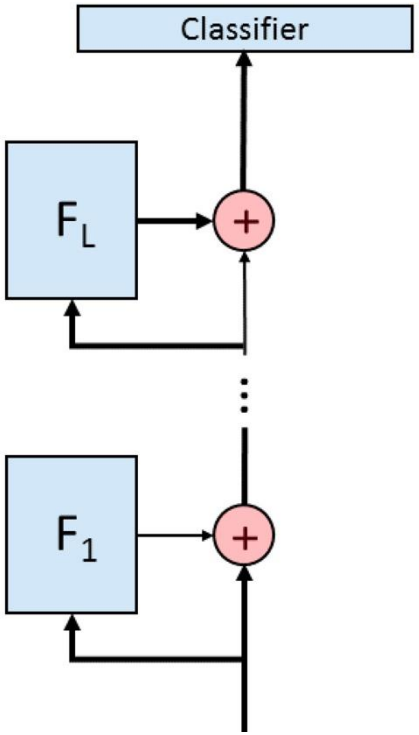
- Insight: To reduce loss, require $F_j(h_j) \propto -\nabla_{h_j} \ell(h_j)$
- This implies that $h_j + F(h_j) \approx h_j - \alpha \nabla_{h_j} \ell(h_j)$
- Refinement of hidden features h_j, not model parameters

Implications

$$\ell(h_L) \approx \ell(h_i) + \underbrace{\sum_{j=i}^{L-1} F_j(h_j) \cdot \nabla_{h_j} \ell(h_j)}$$



$$h_L \approx h_i - \sum_{j=i}^{L-1} \alpha_j \nabla_{h_j} \ell(h_j)$$

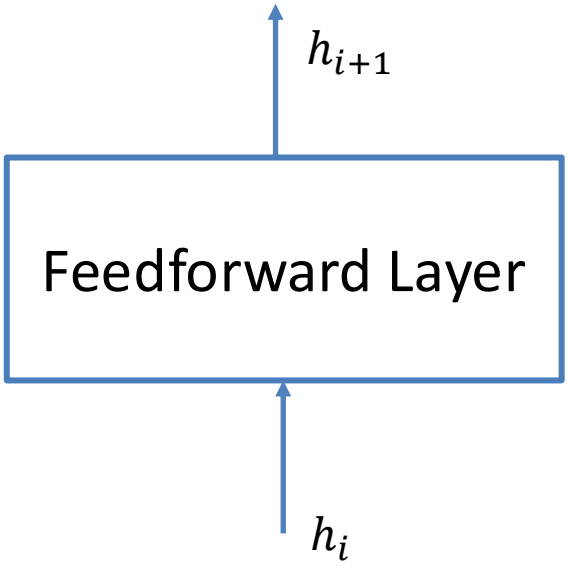


Fascinating Implications

$$h_L \approx h_i - \sum_{j=i}^{L-1} \alpha_j \nabla_{h_j} \ell(h_j)$$

- L residual layers (skip connections) effectively may be viewed as L steps of gradient descent
- That gradient descent is effectively learned within the model
- Data-dependent feature refinement at test time

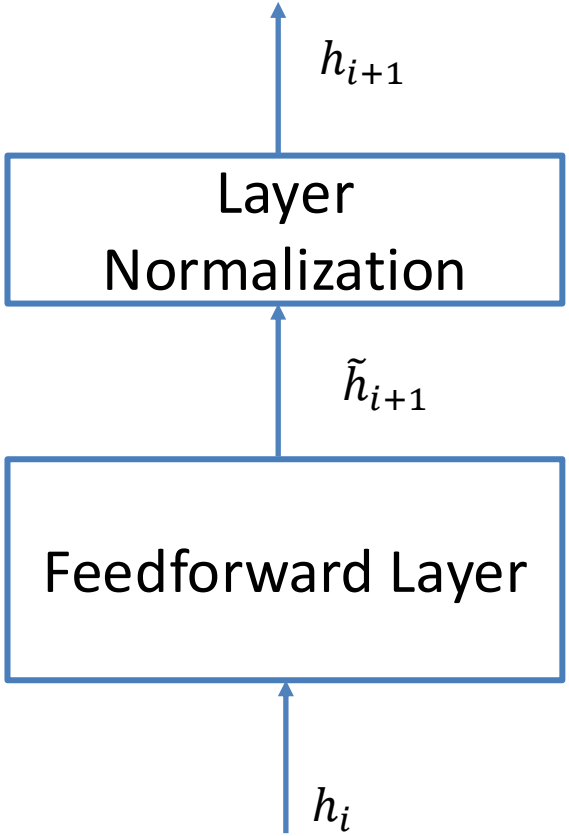
Summary of the Changes Introduced to Deep Networks



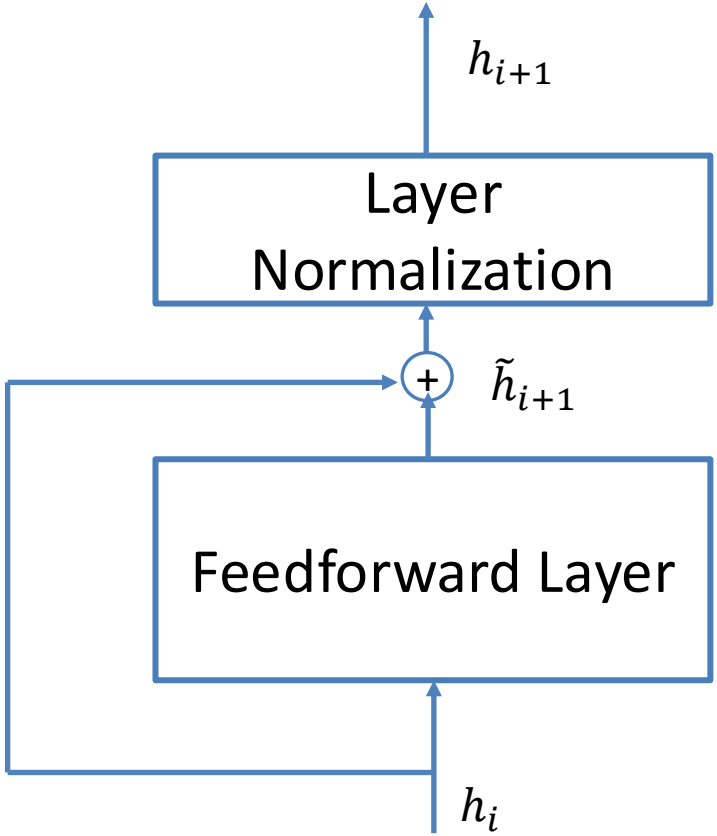
$$h_{i+1} = f(A_i h_i + b_i)$$

$f(\cdot)$ is a pointwise nonlinearity (ReLu)

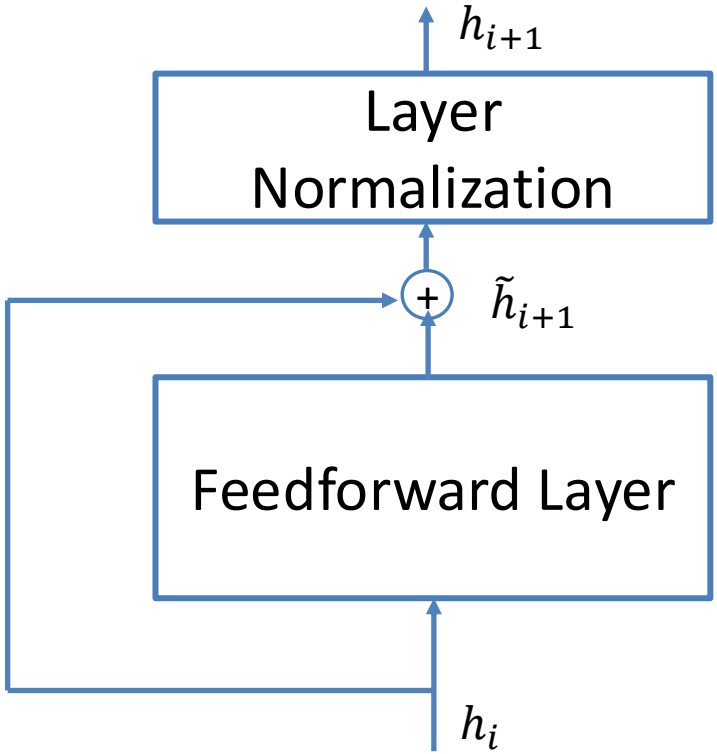
Normalization



Skip Connection

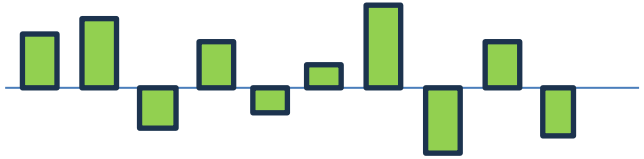


Feedforward Element with Layer Norm & Skip Connection



An Intuitive Explanation of Large Language Models

Vector of
Features



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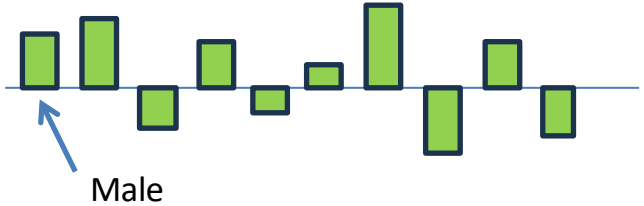

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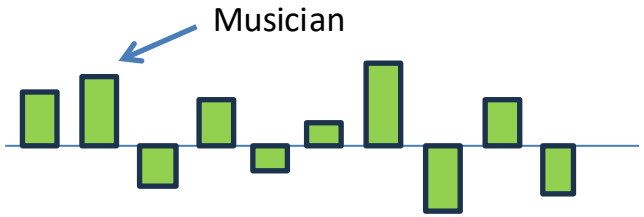

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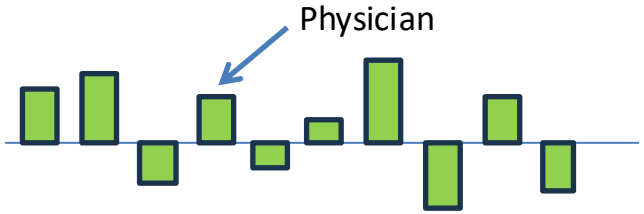

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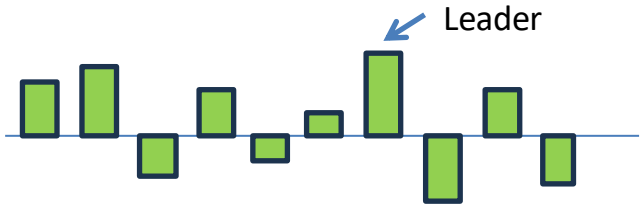

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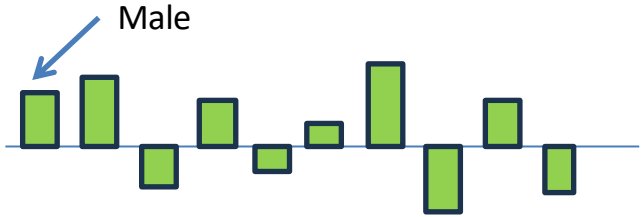
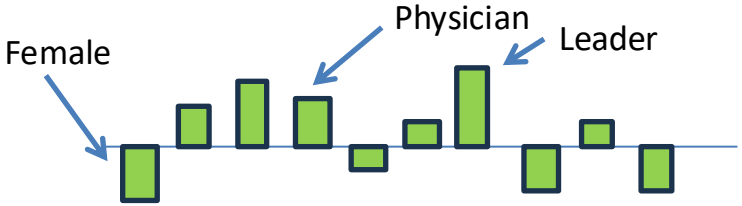
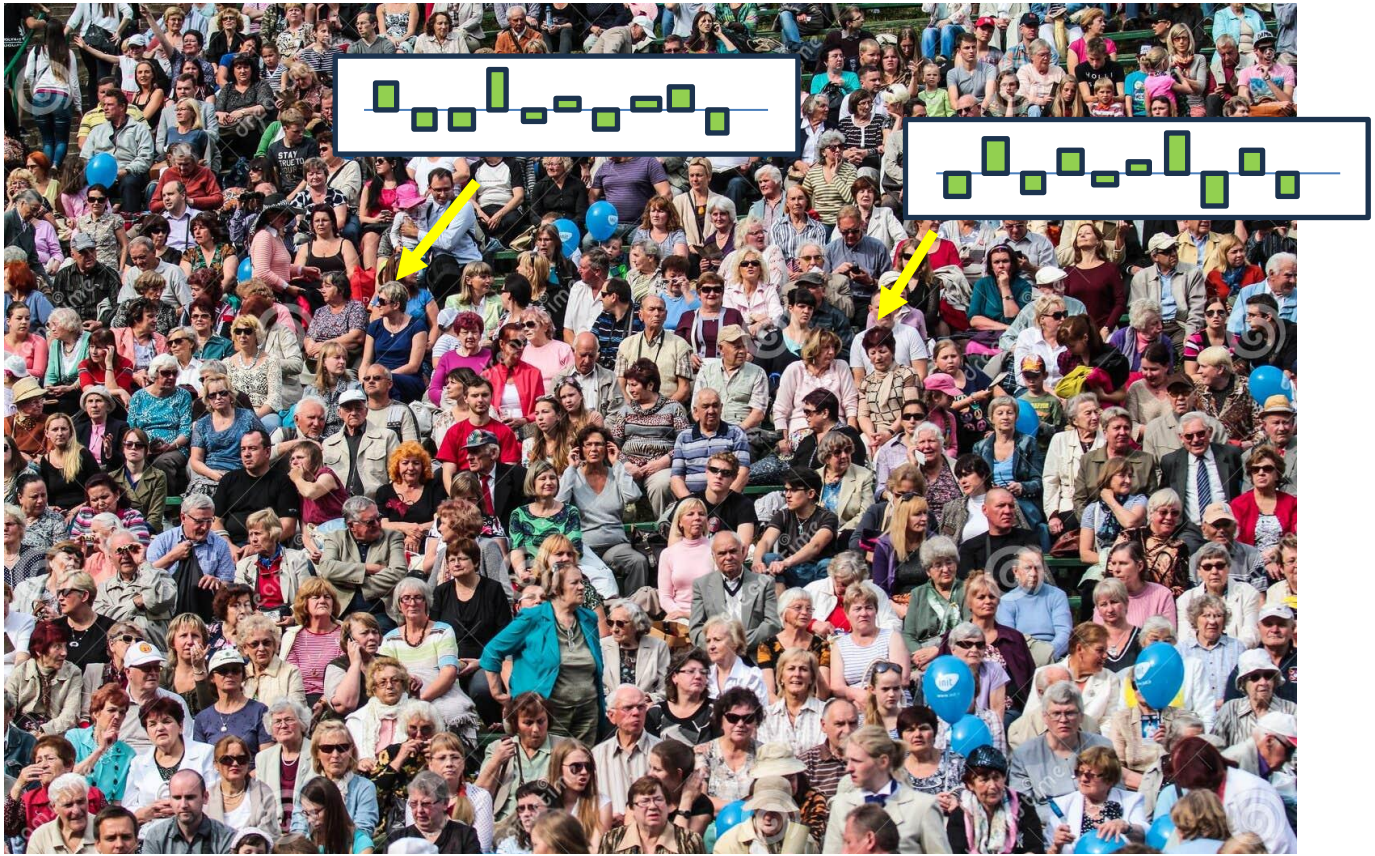
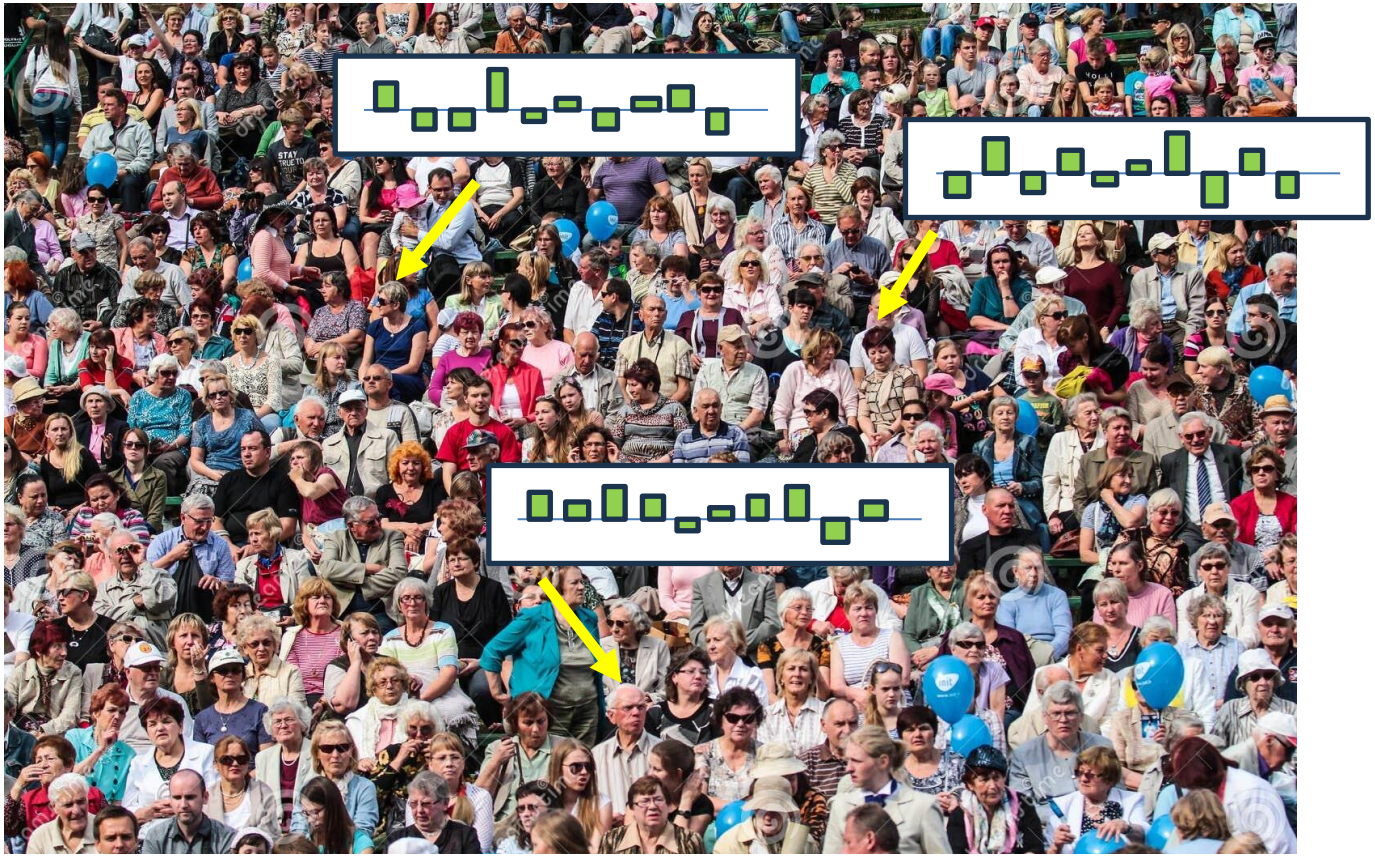


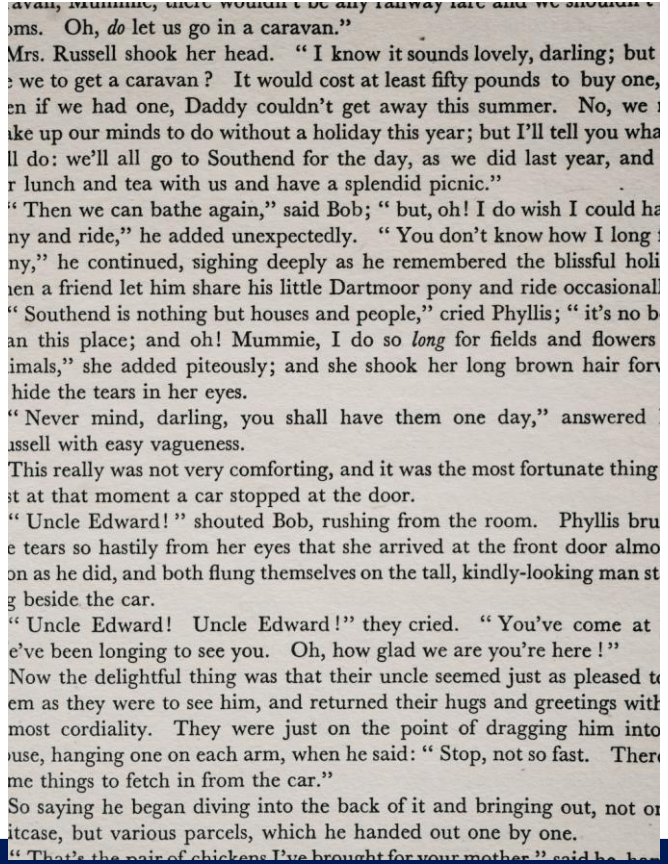






Every Person Characterized by a Vector, Reflecting their Attributes
Strength of each component measured relative to the average across all people

People an Analogy for Words. Every Word Mapped to a Vector of Features



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Mrs. Russell shook her head. "I know it sounds lovely, darling; but we to get a caravan? It would cost at least fifty pounds to buy one, n if we had one, Daddy couldn't get away this summer. No, we ke up our minds to do without a holiday this year; but I'll tell you wha ll do: we'll all go to Southend for the day, as we did last year, and r lunch and tea with us and have a splendid picnic."

"Then we can bathe again," said Bob; "but, oh! I do wish I could ha ny and ride," he added unexpectedly. "You don't know how I long ny," he continued, sighing deeply as he remembered the blissful holi en a friend let him share his little Dartmoor pony and ride occasionall

"Southend is nothing but houses and people," cried Phyllis; "it's no b an this place; and oh! Mumme, I do so *long* for fields and flowers imals," she added piteously; and she shook her long brown hair forv hide the tears in her eyes.

"Never mind, darling, you shall have them one day," answered ssell with easy vagueness.

This really was not very comforting, and it was the most fortunate thing st at that moment a car stopped at the door.

"Uncle Edward!" shouted Bob, rushing from the room. Phyllis bru e tears so hastily from her eyes that she arrived at the front door almo on as he did, and both flung themselves on the tall, kindly-looking man st g beside the car.

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Now the delightful thing was that their uncle seemed just as pleased t em as they were to see him, and returned their hugs and greetings wit most cordiality. They were just on the point of dragging him into use, hanging one on each arm, when he said: "Stop, not so fast. Ther me things to fetch in from the car."

So saying he began diving into the back of it and bringing out, not o itcase, but various parcels, which he handed out one by one.

"That's the pair of chickens I've brought for your mother," said he, ho

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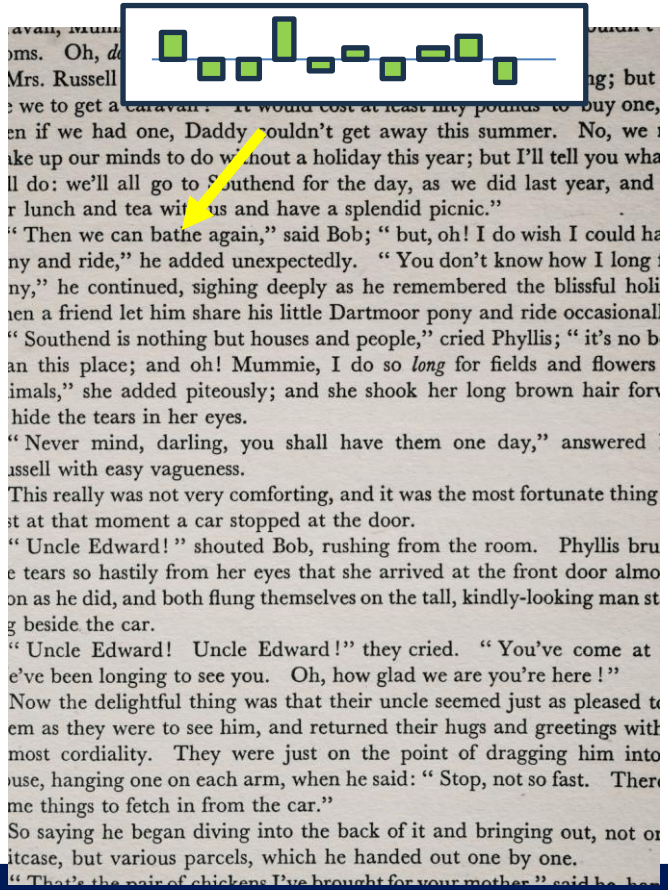

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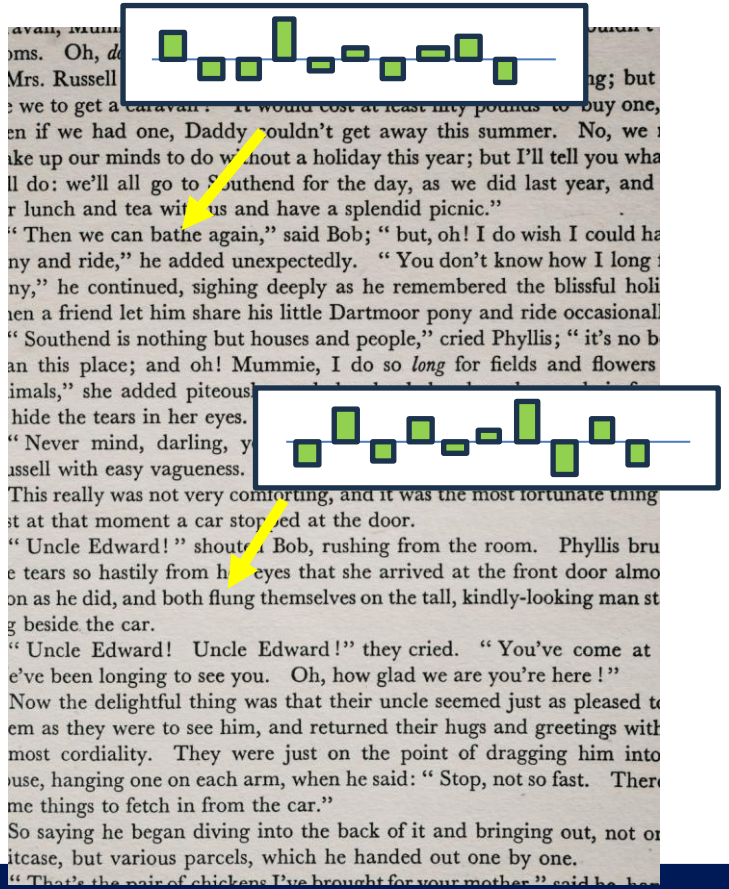

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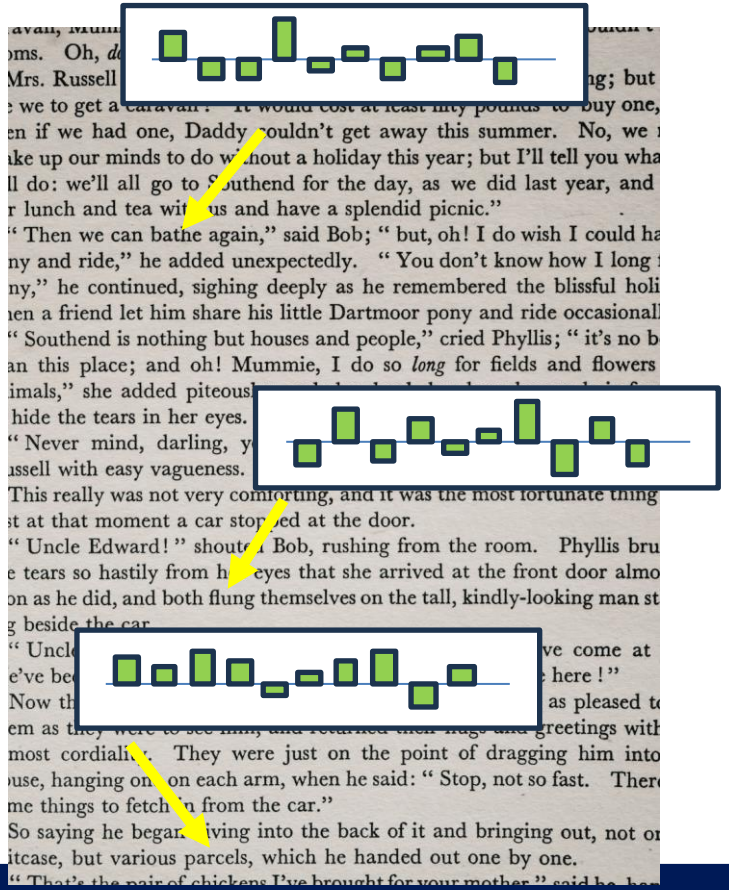

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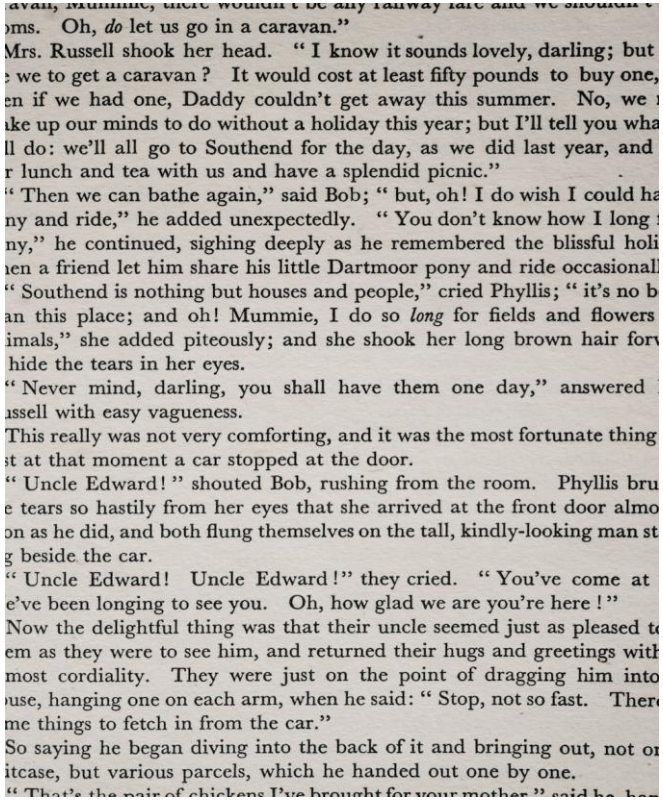

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Every Word Characterized by a Vector, Reflecting their Attributes/Definition
Strength of each component measured relative to the average across all words

An Analogy to Explain Large Language Models

- Each person (word) is characterized by a set of features

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- Features could even be self-contradictory

  - *Cleave*: “split or sever something,” “adhere very strongly to”

An Analogy to Explain Large Language Models

- Each person (word) is characterized by a set of features

- Features could even be self-contradictory

  - *Cleave*: “split or sever something,” “adhere very strongly to”

- Context is important:

  - Which features are relevant dictated by the context: the surrounding people (words)
  - Would like to develop a method that can adjust the features of a person based on context
  - Use here the intuition of people represented by a set of features to motivate NLP framework
  - In NLP, words represented by vectors, with features associated with different meanings
  - Words have multiple meanings (features), with the relevance of each dependent on context

- People (words) that surround each other likely to have some features in common, define context
- Reinforce those features that are common among nearby people (words)
- Emphasize features based on context

How Do We Algorithmically Account for Context?

- ❑ Learn a baseline, context-independent feature vector for each person (word)
- ❑ Revise the baseline vector based on the context of the surrounding people (words)
- ❑ The more similar the surrounding vectors to the baseline vector of the person (word) of interest, the more relevant is it for context
- ❑ The more physically proximate/nearby the person (word), the more relevant the context

- The highlighted people may have very diverse interests and backgrounds, but because they are all together, the common element of being a physician should be highlighted

- The *type* of clinicians provides context on health condition of patient

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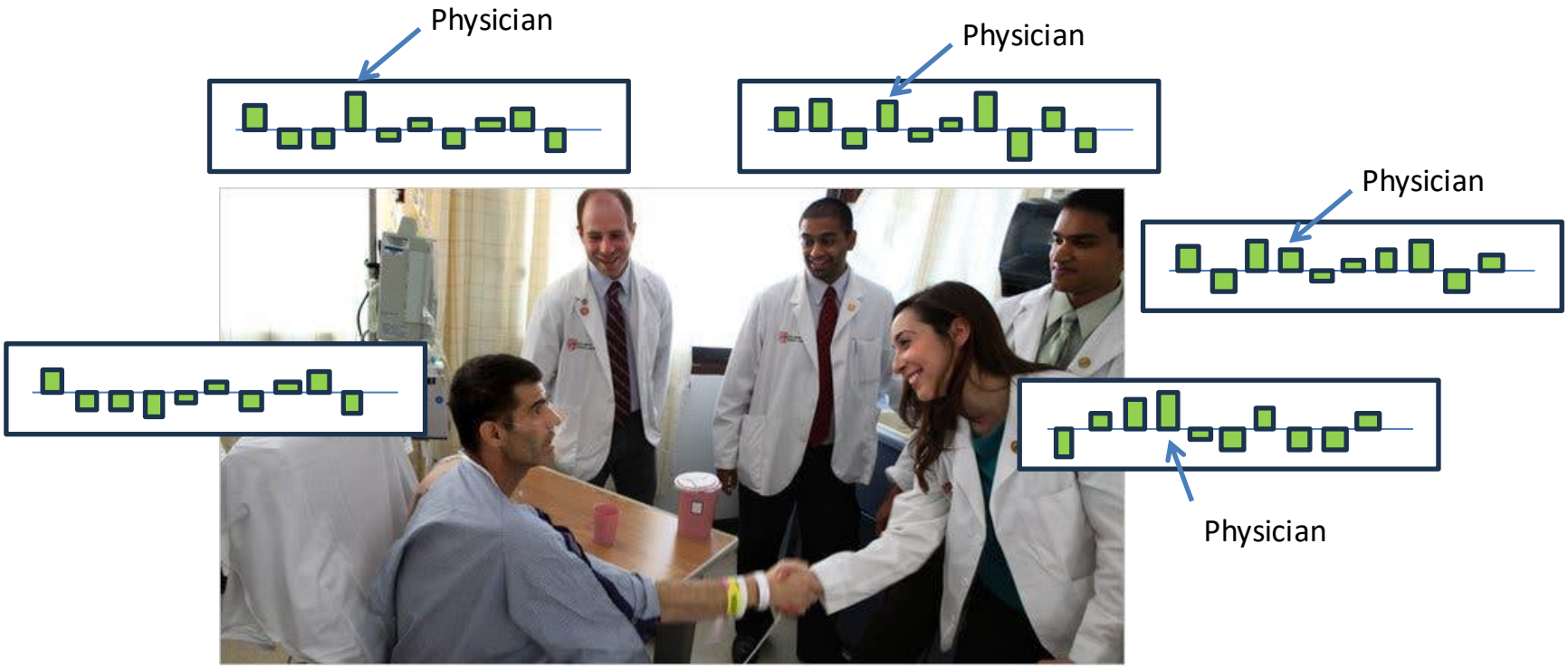
- Baseline (normal) features of the patient likely less relevant to context

- The highlighted people may have very diverse interests and backgrounds, but because they are all together, the common element of being a physician should be highlighted
- The type of clinicians provides context on health condition of patient
- Baseline (normal) features of the patient likely less relevant to context
- Presence of multiple nearby doctors, with likely shared specialty, tells us updated information about patient, based on context

Initial (Baseline) Feature Vectors for Each Person



For Each Person, Update Vectors As Weighted Sum of *All* Vectors

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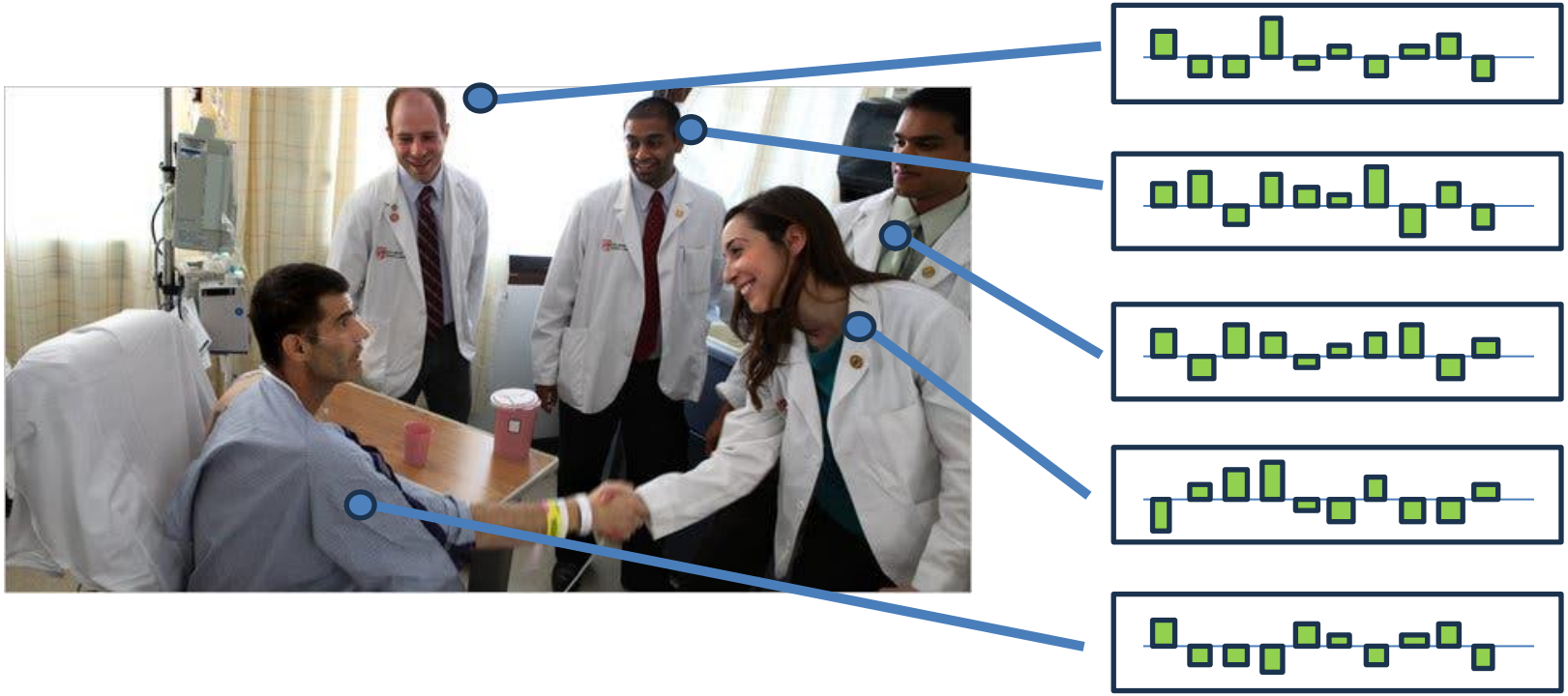
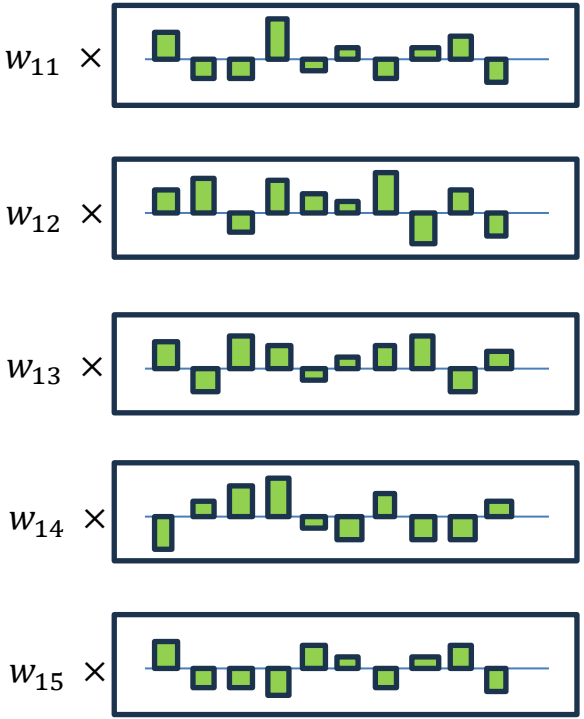


For Each Person, Update Vectors As Weighted Sum of All Vectors

Person 1



$$w_{11} + w_{12} + w_{13} + w_{14} + w_{15} = 1$$

For Each Person, Update Vectors As Weighted Sum of All Vectors

Person 1

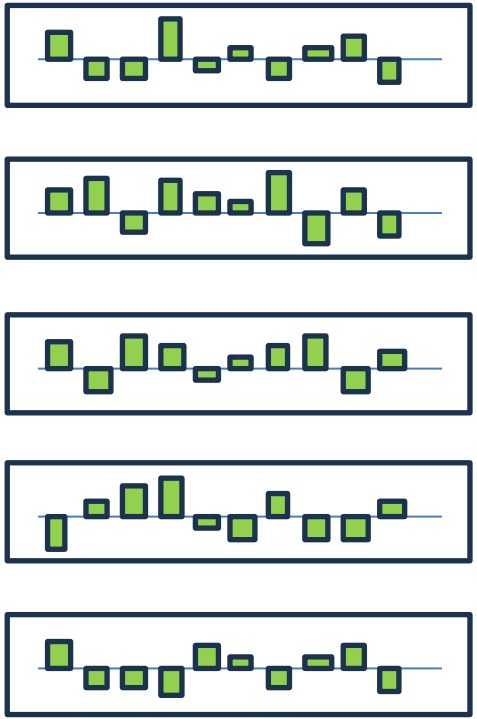
$w_{11} \times$

$w_{12} \times$

$w_{13} \times$

$w_{14} \times$

$w_{15} \times$

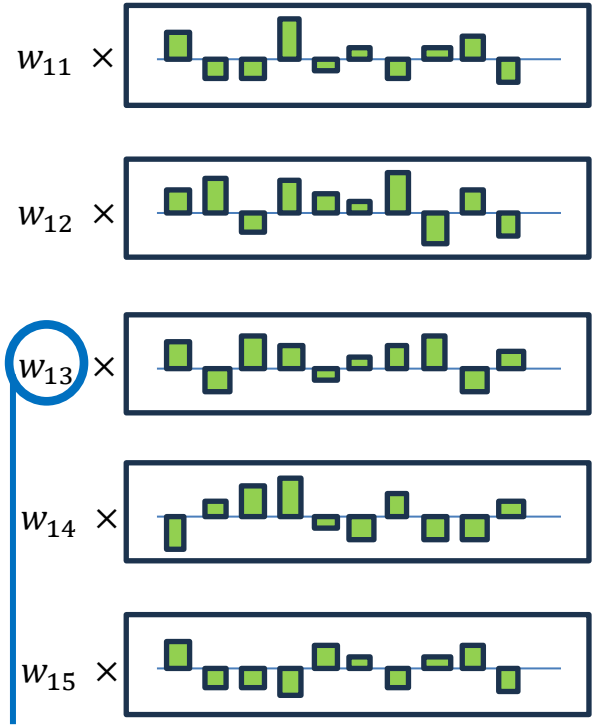


$$w_{11} + w_{12} + w_{13} + w_{14} + w_{15} = 1 \qquad w_{1j} > 0$$

For Each Person, Update Vectors As Weighted Sum of All Vectors

Person 1

$$w_{11} + w_{12} + w_{13} + w_{14} + w_{15} = 1$$



Relative relevance of Person 3 to Person 1

For Each Person, Update Vectors As Weighted Sum of All Vectors

$$w_{11} + w_{12} + w_{13} + w_{14} + w_{15} = 1$$

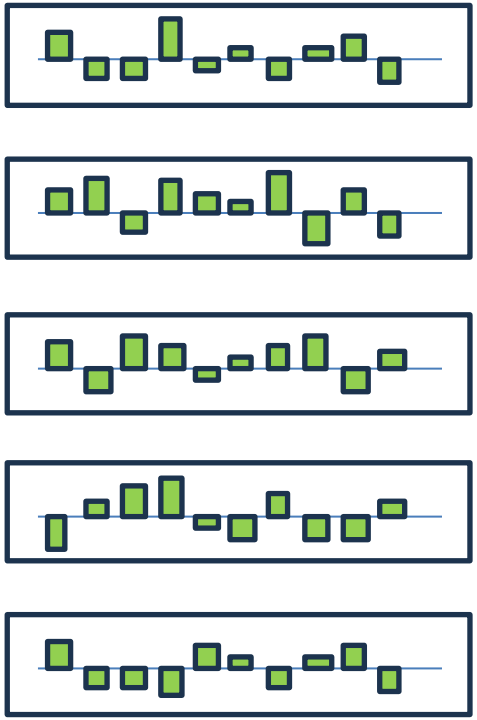
$w_{11} \times$

$w_{12} \times$

$w_{13} \times$

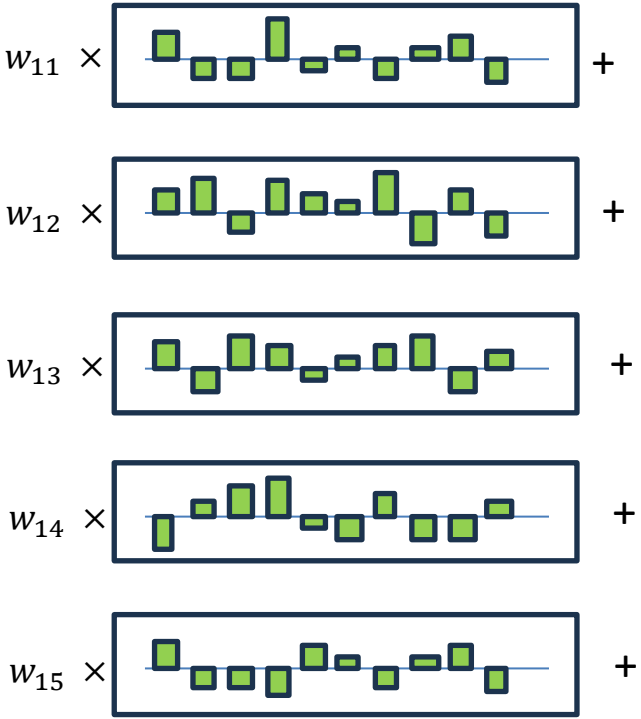
$w_{14} \times$

$w_{15} \times$



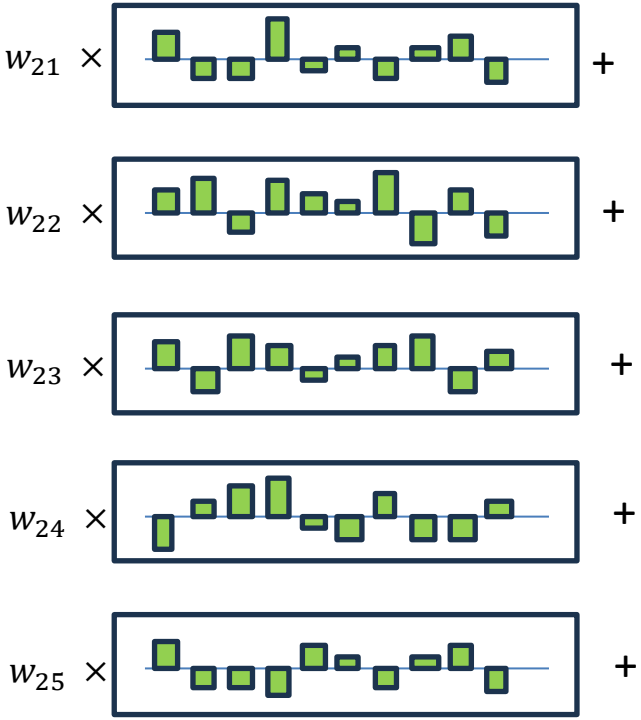
Weights determined by similarity of vectors to Person 1 vector, and physical proximity

For Each Person, Update Vectors As Weighted Sum of All Vectors



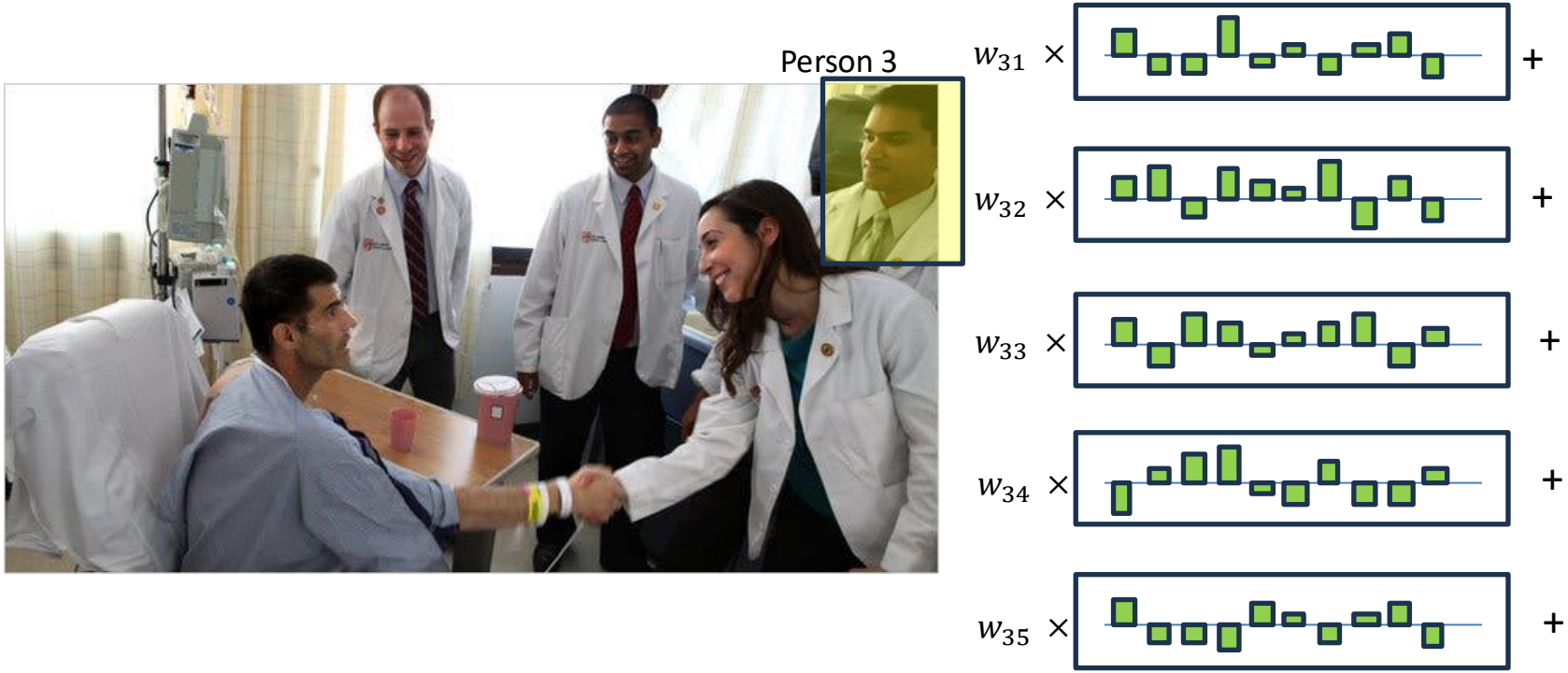
Update Feature Vector for Person 1

For Each Person, Update Vectors As Weighted Sum of All Vectors

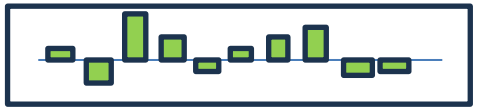


Update Feature Vector for Person 2

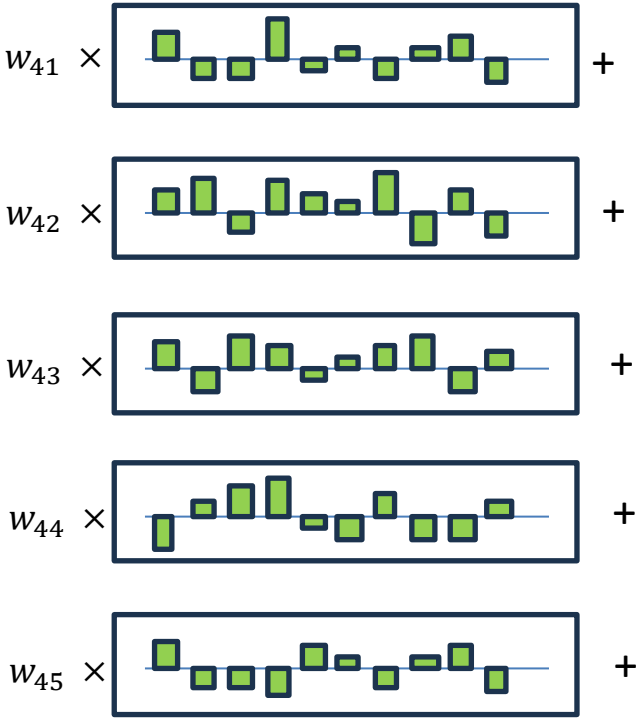
For Each Person, Update Vectors As Weighted Sum of All Vectors



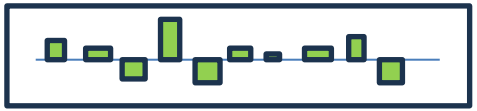
Update Feature Vector for Person 3

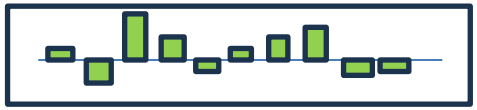


For Each Person, Update Vectors As Weighted Sum of All Vectors

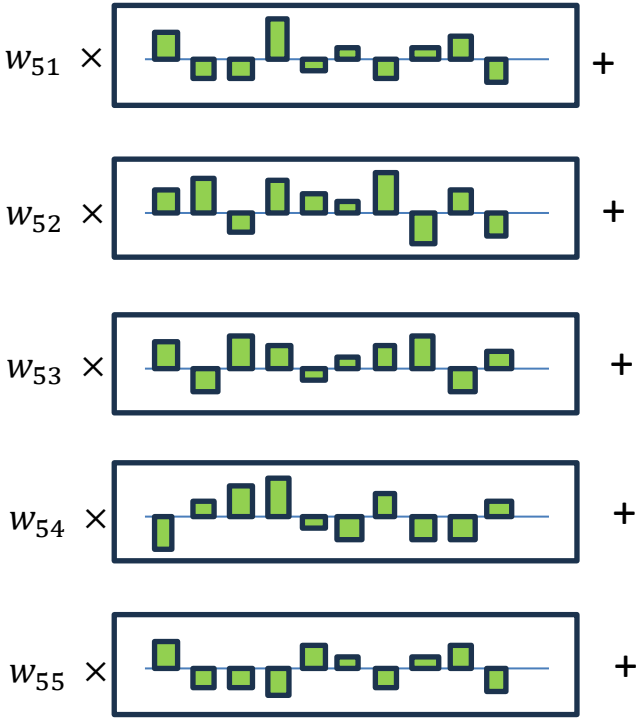


Update Feature Vector for Person 4

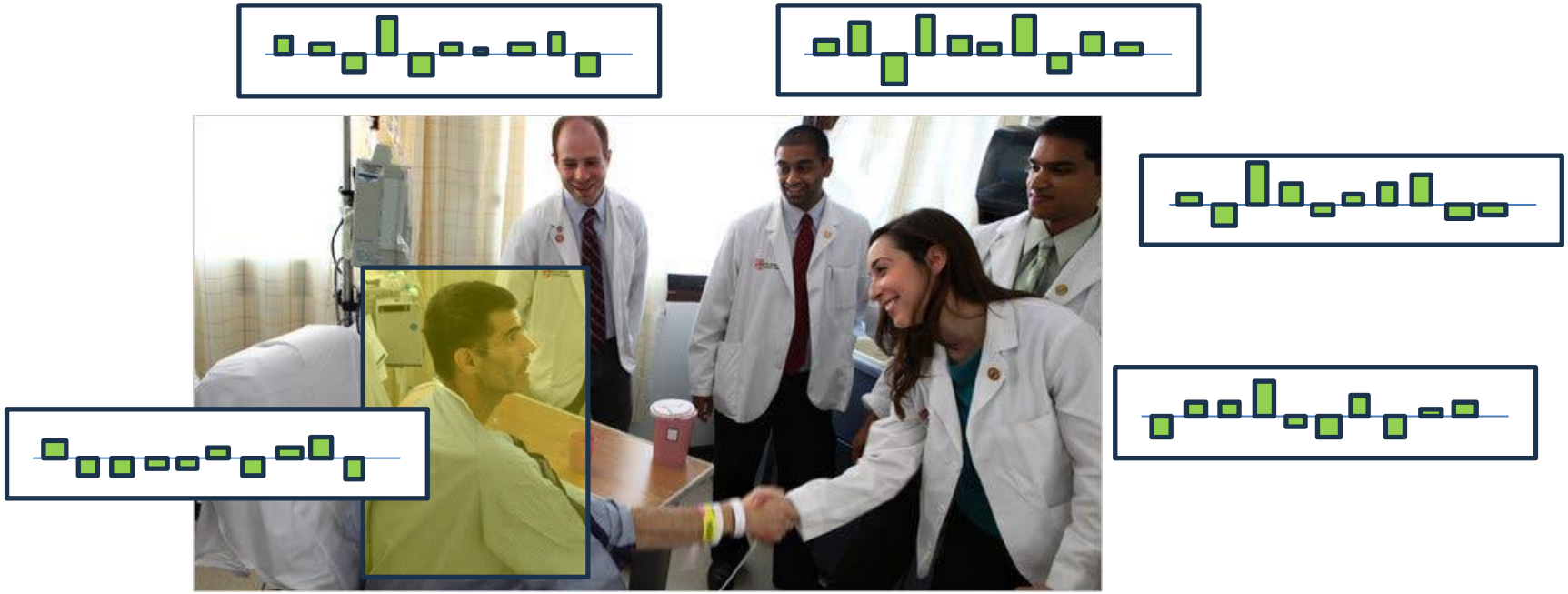




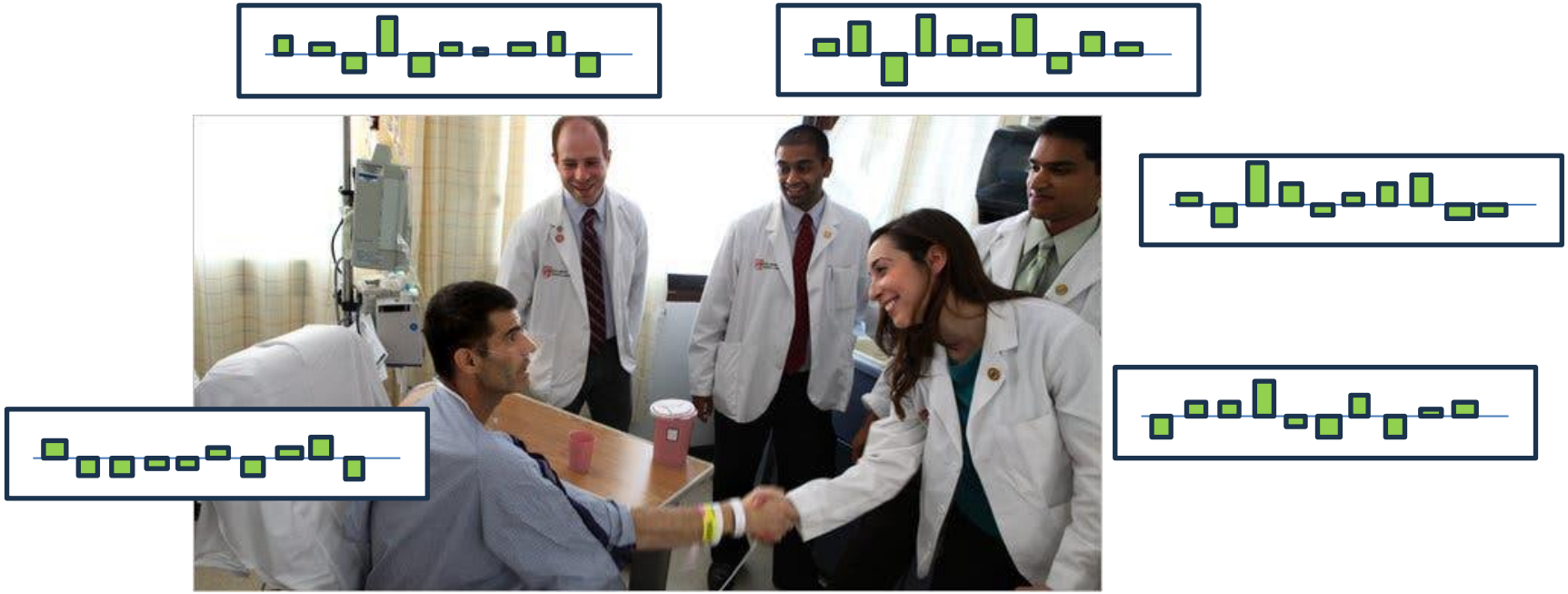
For Each Person, Update Vectors As Weighted Sum of All Vectors



Update Feature Vector for Person 5



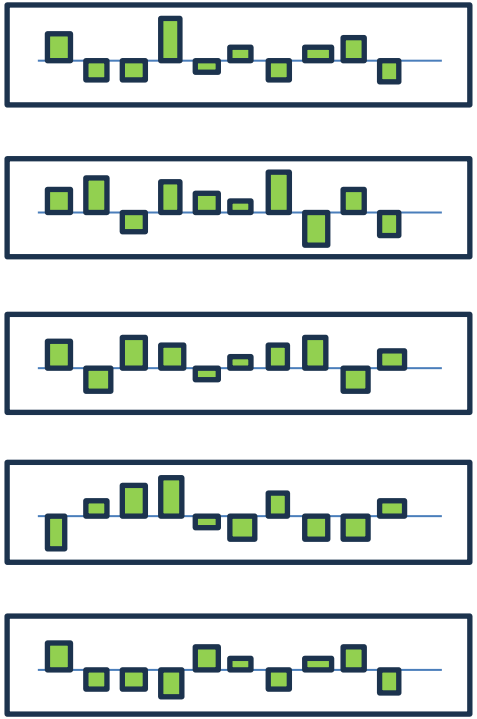
Update Features for All People

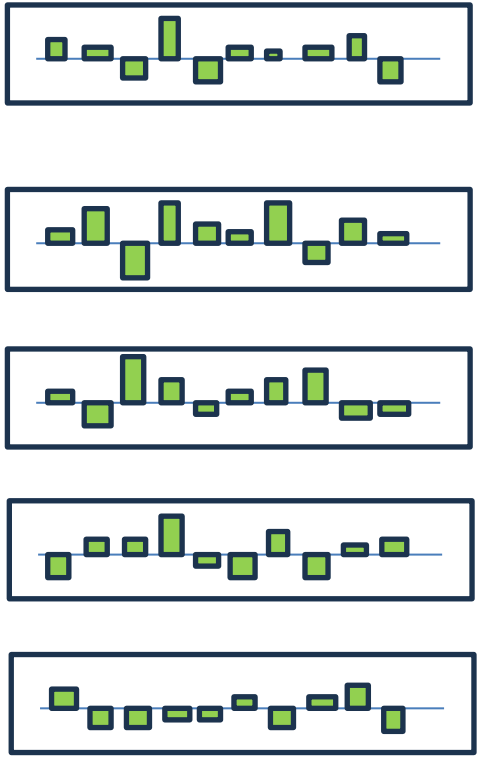


- The presence of four surrounding MDs reinforces this is a medical situation
- Idiosyncratic features of the MDs *not* reinforced; commonality (specialty) is emphasized in updated vectors
- Whatever the characteristics of the patient initially, his features are now influenced by the presence of four MDs

- Same five people, situated in <u>this</u> scene, would have very different updated contextual vectors
- In this scene, updated feature vectors may emphasize that they all went to same university, or that they were all sports fans

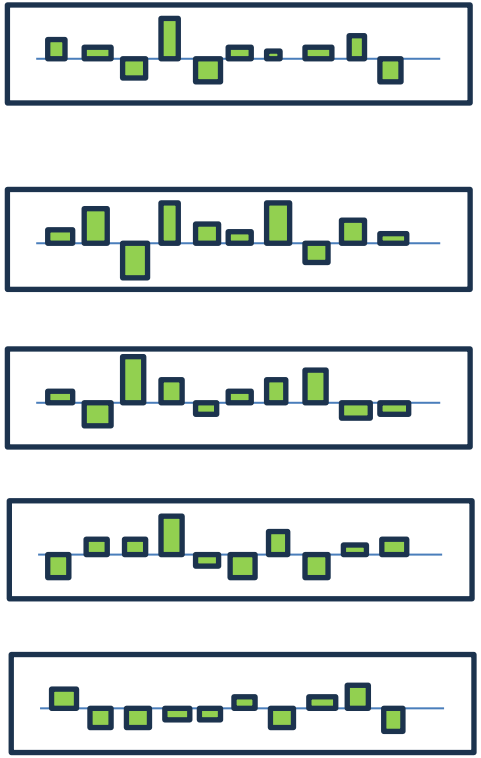
Viewed as a Functional Operation





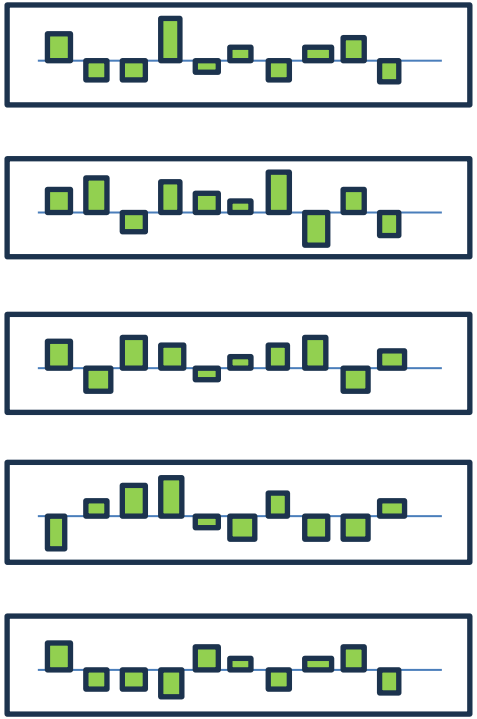
Viewed as a Functional Operation

Input vectors, context independent



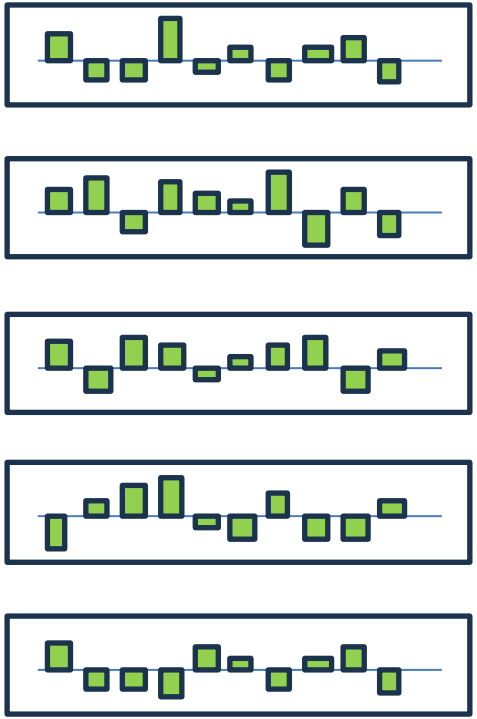
Viewed as a Functional Operation

Output vectors, context dependent

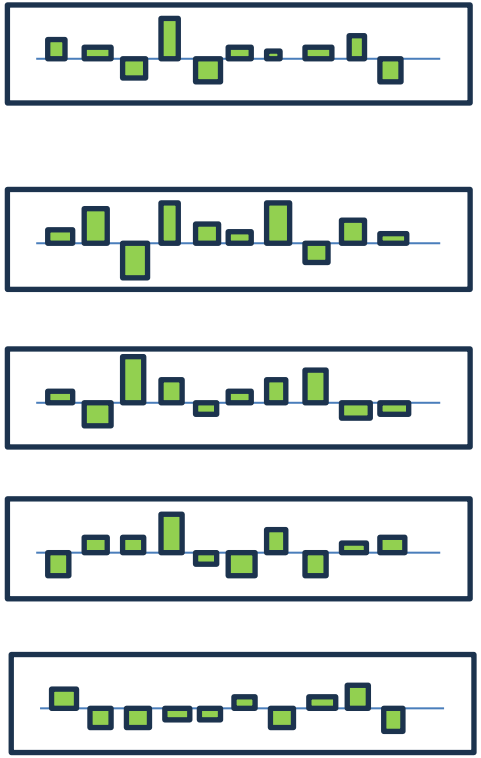


Viewed as a Functional Operation

Input

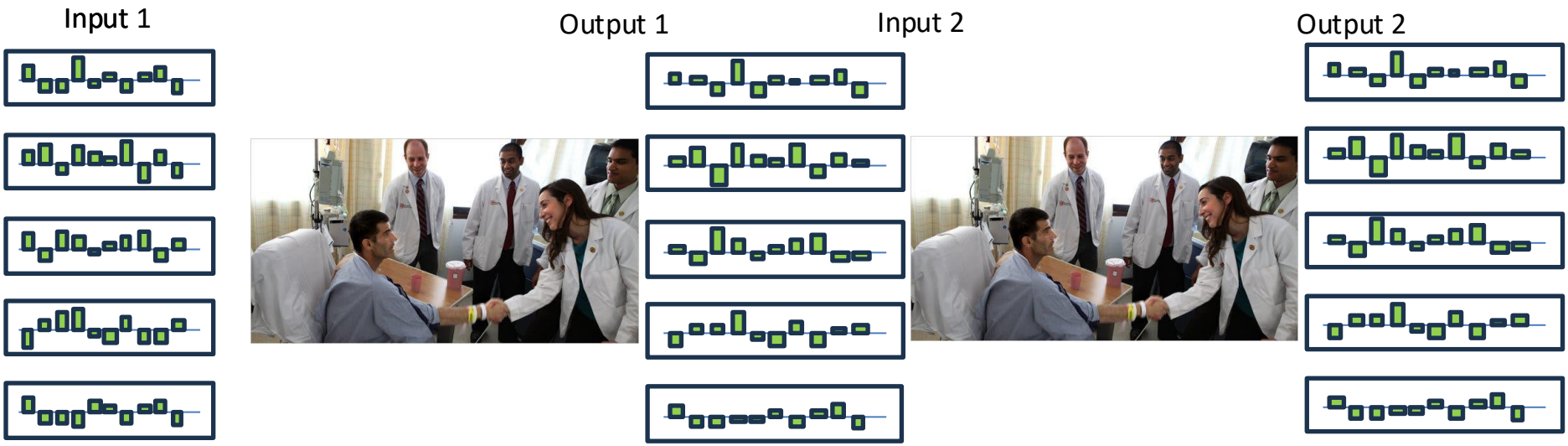


Output



Functional Relationship Between Input & Output Vectors Defined by Context of Full Scene

Now that Vectors Updated, <u>Repeat on Updated Vectors</u>, to Further Exploit Context



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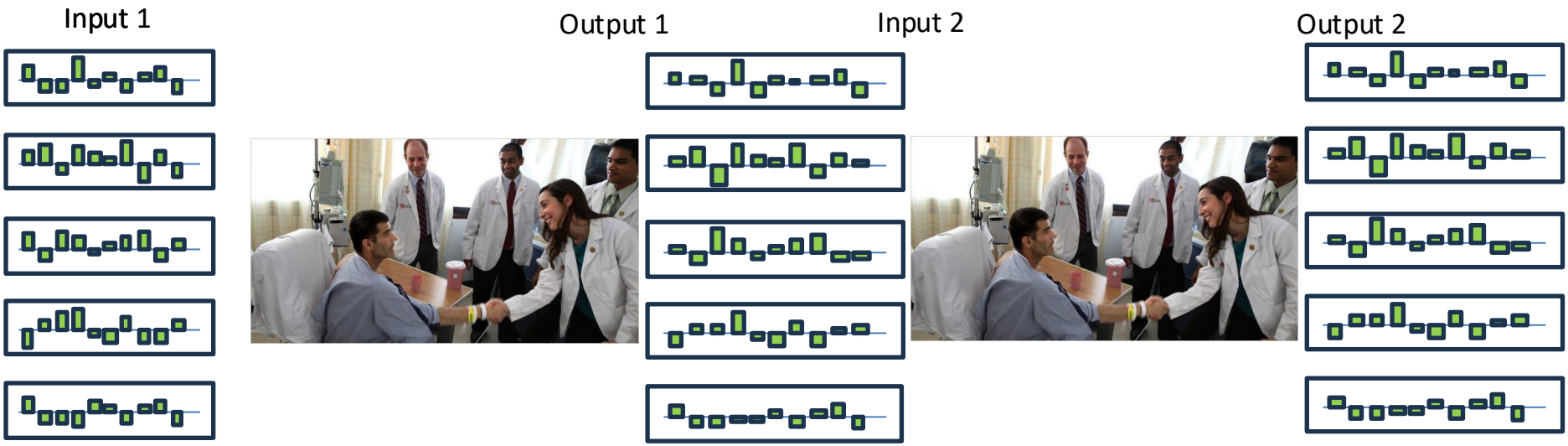

Repeat *L* times, for an *L*-layer model: **Deep Learning**

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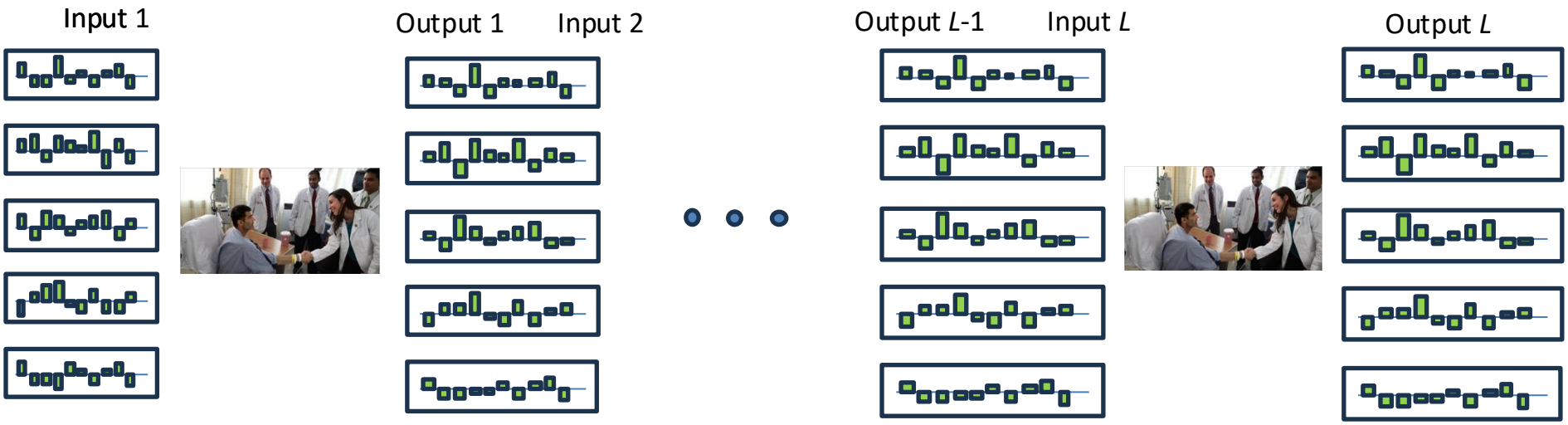

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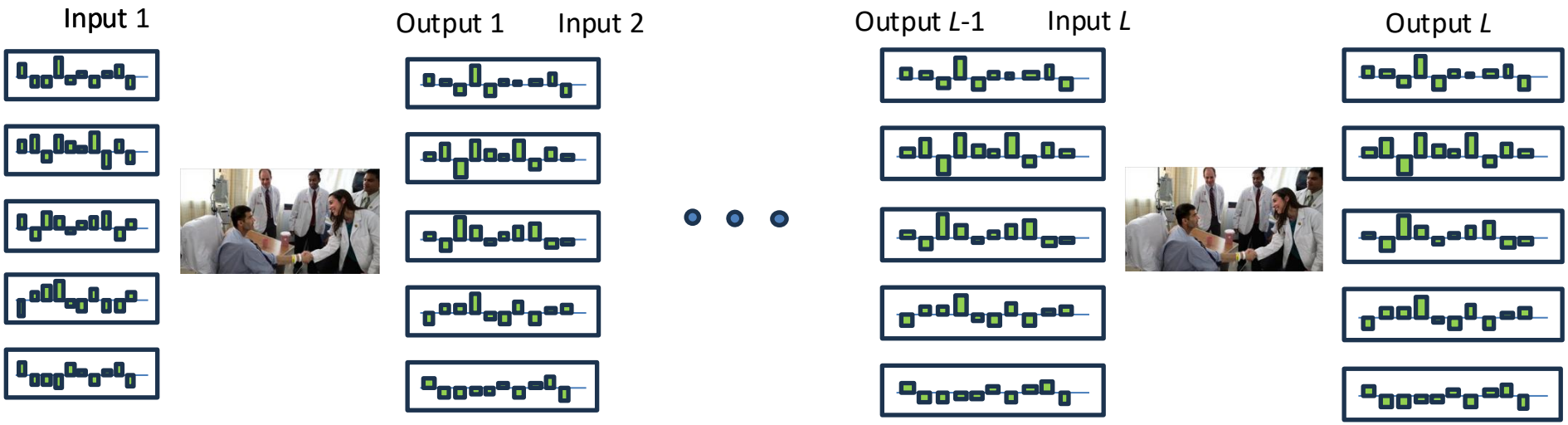
Prediction Task



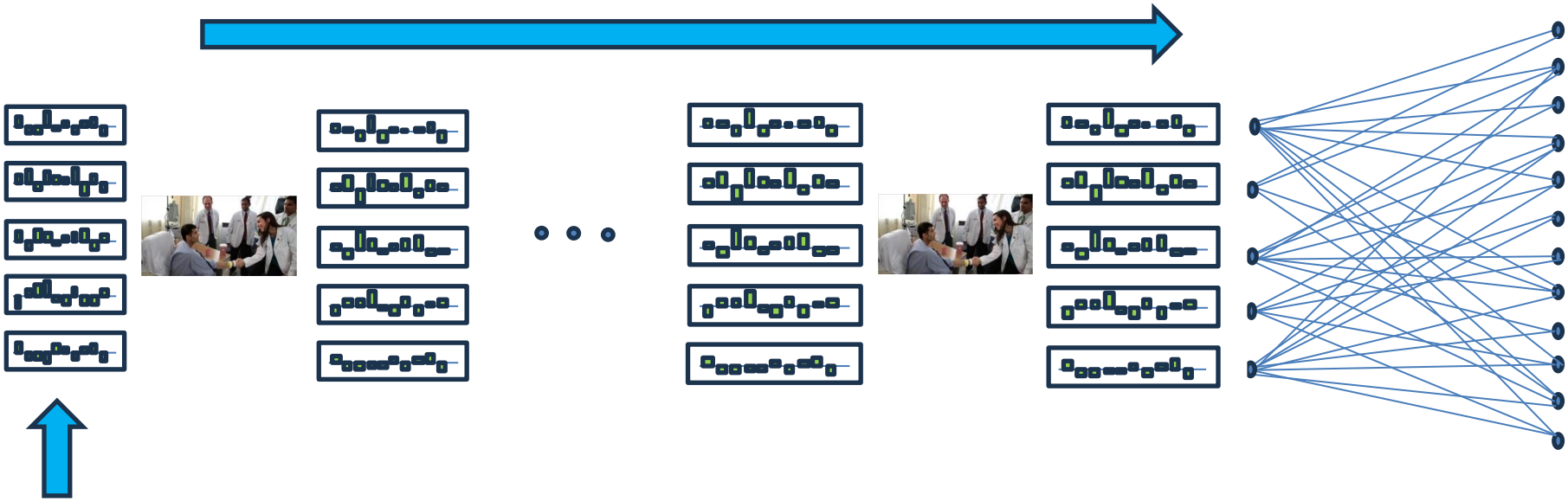
- Given the feature vectors output from Layer L, prediction task: Who is the next person to enter the room?
- Relative? Doctor? Nurse? Dancer? Businessman? Waiter? ….

Transformer Network

Context-independent vector for each word (“person”)

Transformer Network

L layers of context-imposition, attention defined by word (“person”) similarity & proximity



Context-independent vector for each word (“person”)

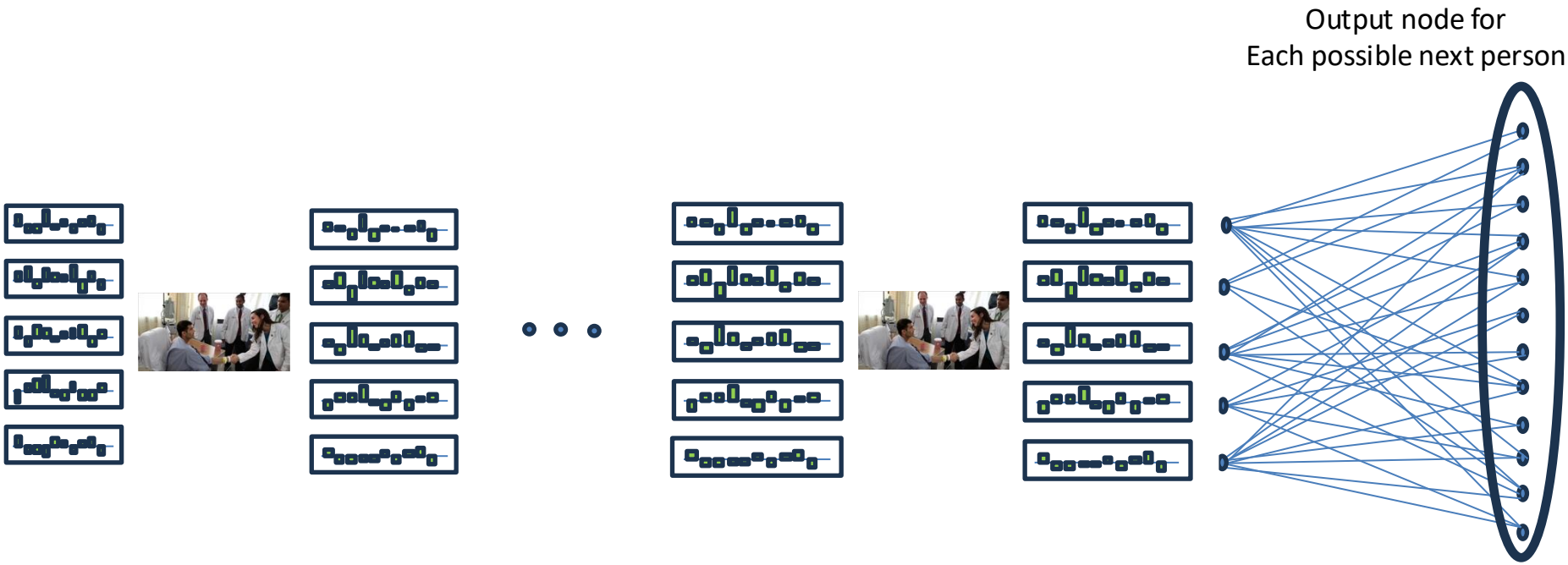
Transformer Network

L layers of context-imposition, defined by word (“person”) similarity & proximity

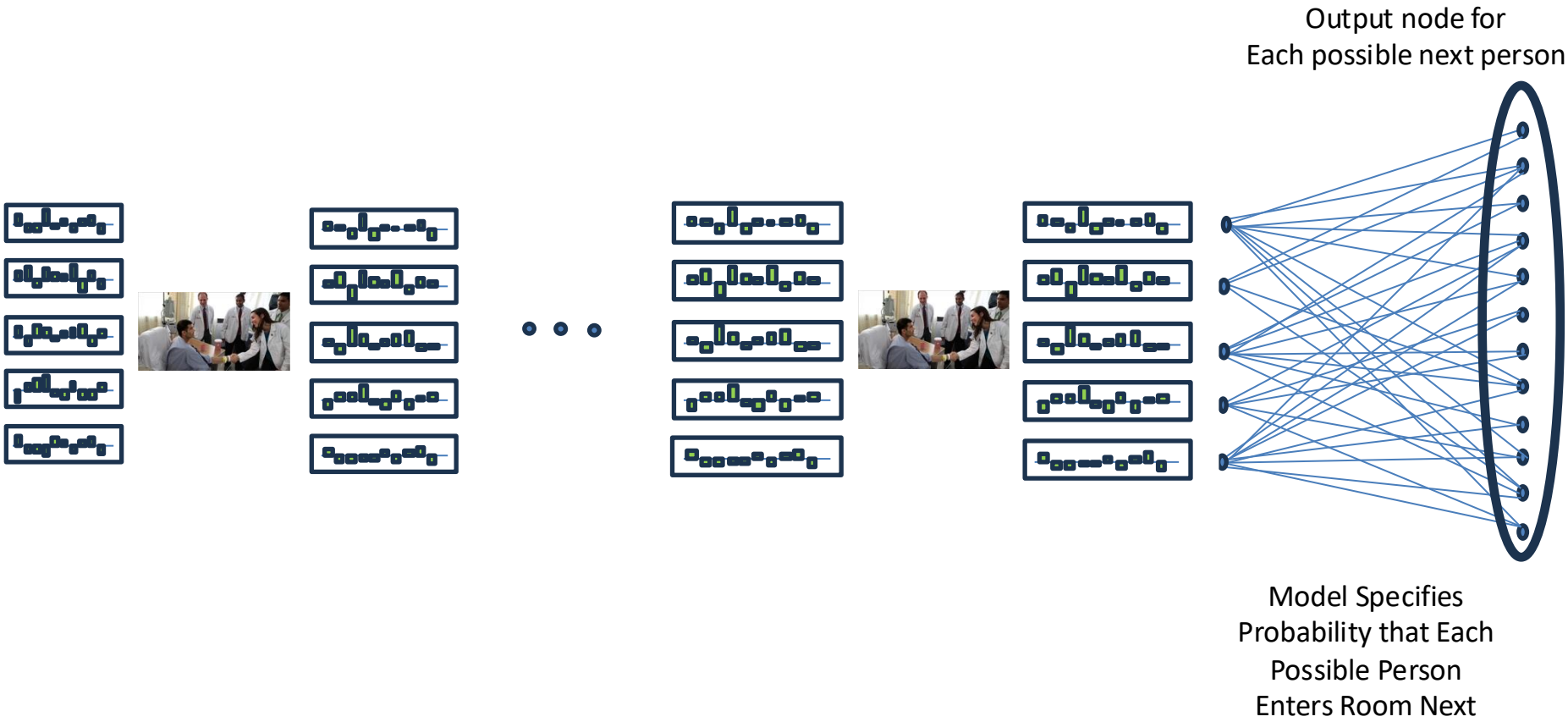
Context-independent vector for each word (“person”)

Predict next word (“person”)

Transformer Network



Transformer Network



Generative Pre-trained Transformers (GPT)

- Rather than building a separate model for each of these tasks, with specialized training data, we first build a universal model based on as much text data as possible, across all disciplines and languages

- This is termed “pre-training” as the resulting model can be refined and applied to many subsequent tasks

- Prediction task: The model should “generate” the next word, given the context of a block of text

- Based upon the “transformer” construct develop above, with people as analogy for words

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- Hence name: **Generative Pre-trained Transformer (GPT)**

Generative Pre-trained Transformers (GPT)

We have A LOT of documents to train on!

Generative Pre-trained Transformers (GPT)

Repeatedly, across all segments of text:
Predict next word

Generative Pre-trained Transformers (GPT)

- We have access to an almost unlimited quantity of documents, in all languages

- We may have many specific tasks we'd like to do with an NLP model ("fine tuning"):

  - Question and answers

  - Draft letters and email response

  - Summarization

  - Translation

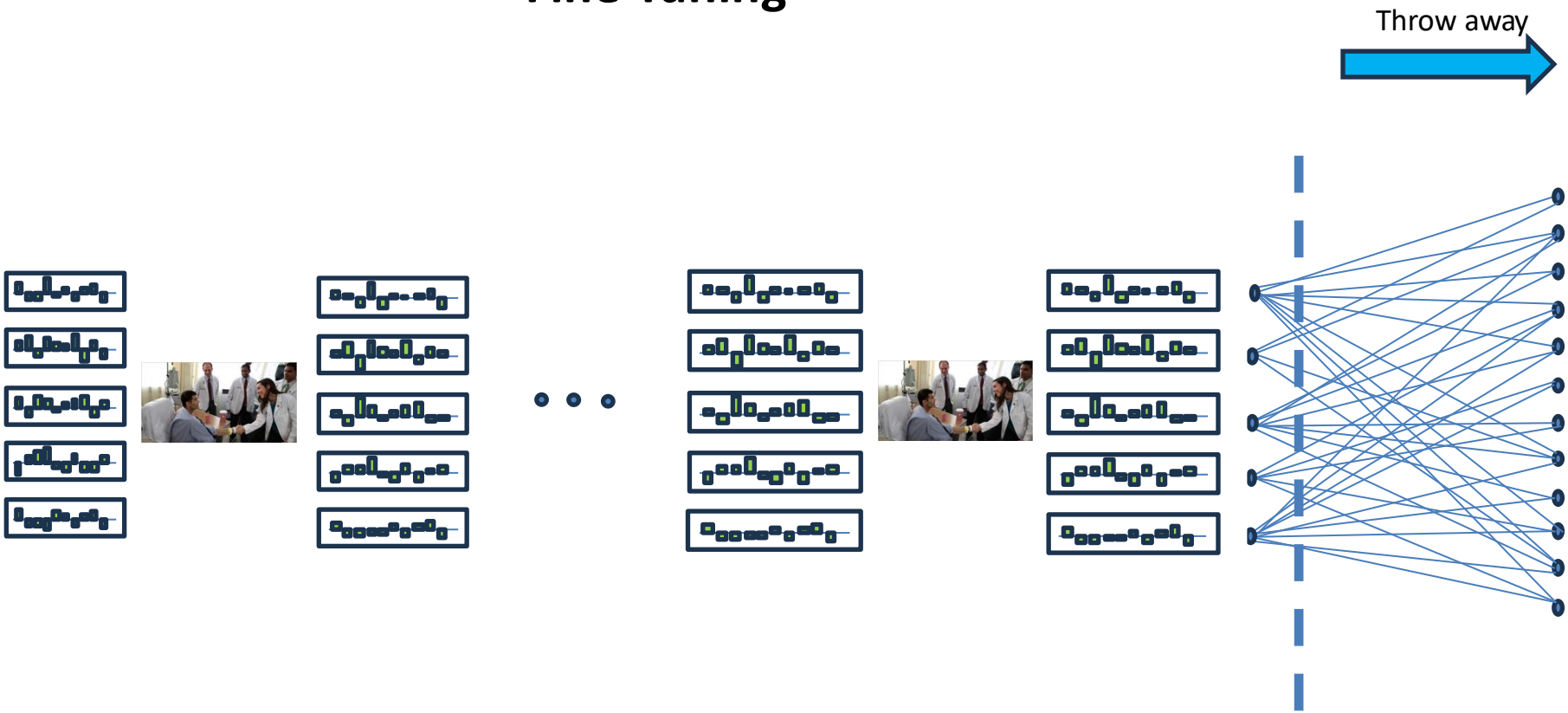
Generative Pre-trained Transformer (GPT)

L layers of context-imposition, defined by word (“person”) similarity & proximity

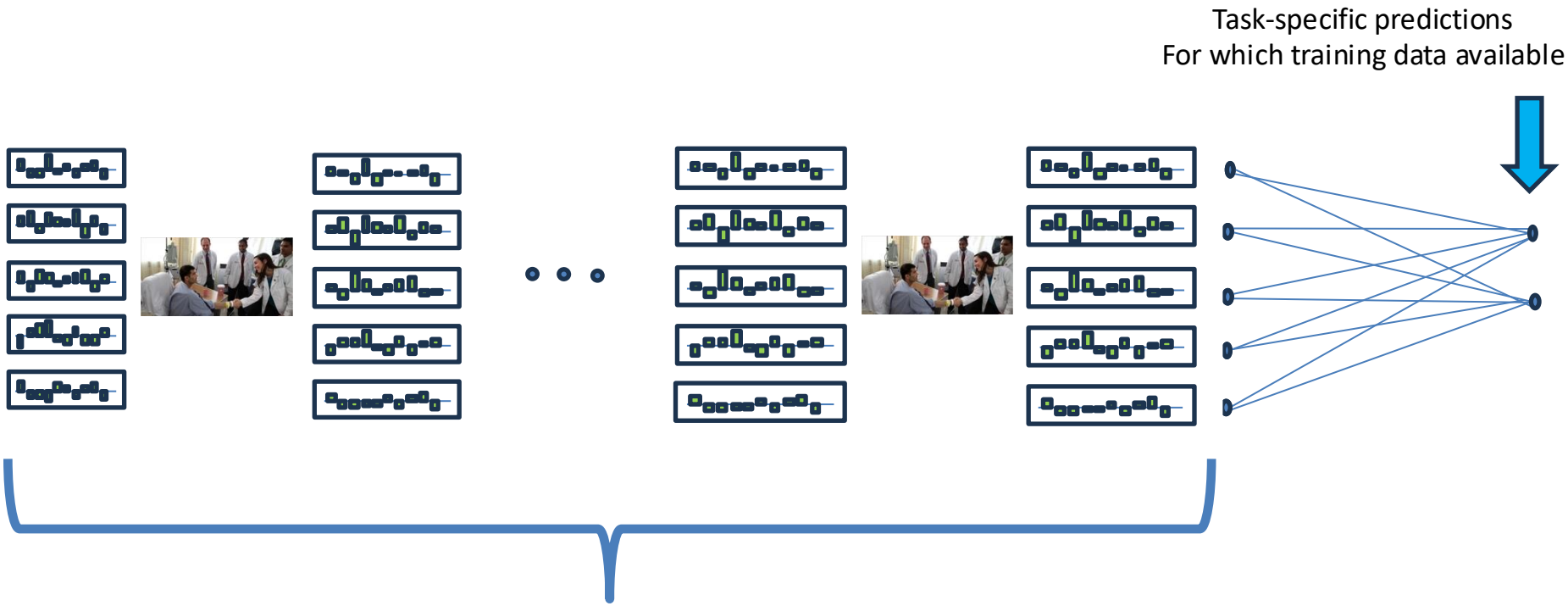
Context-independent vector for each word (“person”)

Predict next word (“person”)

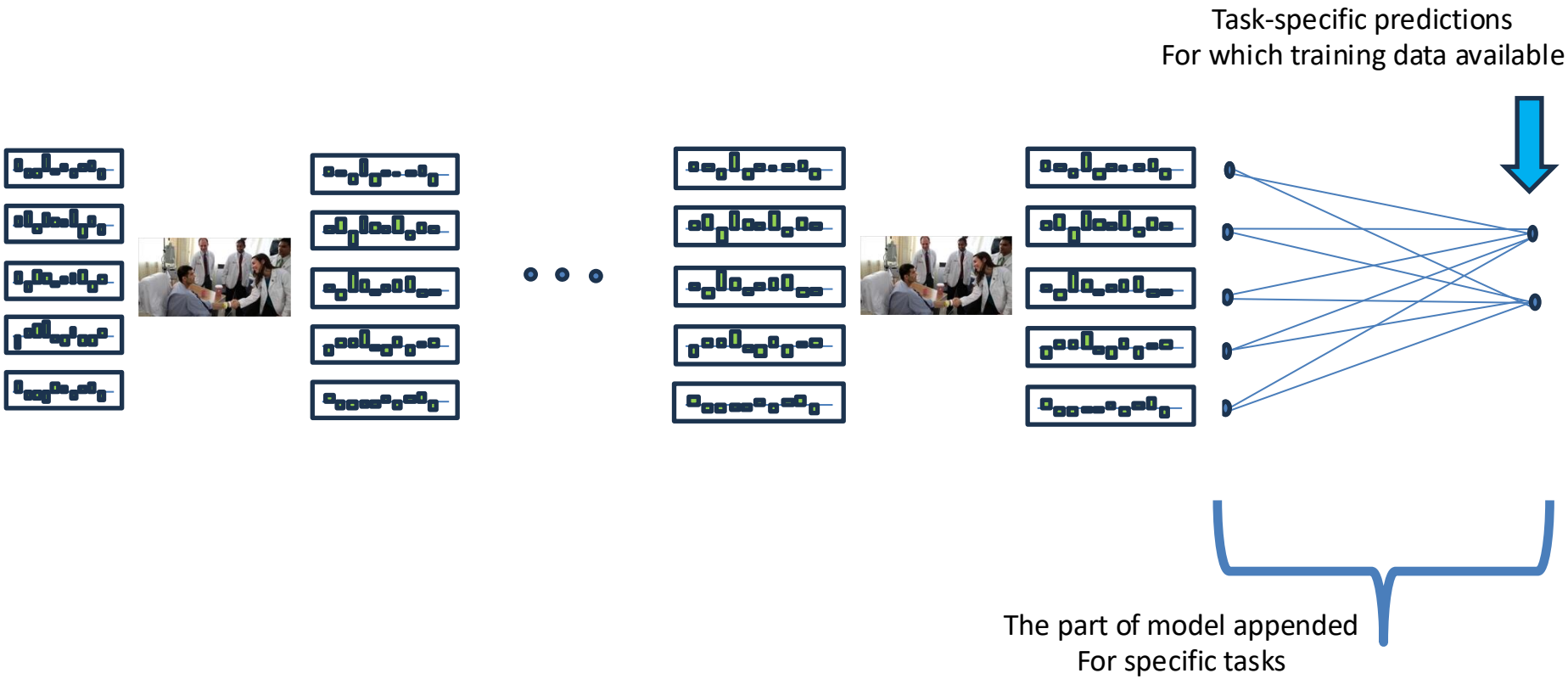
Fine Tuning



Fine Tuning


Task-specific predictions
For which training data available
Fine-tune/refine/update parameters from GPT for
desired specific prediction tasks

Fine Tuning


Task-specific predictions
For which training data available
The part of model appended
For specific tasks

Diving into Details of LLMs